EMBER: An Interactive Database Quantifying the Economic and Environmental Impacts of Wildfires on Western U.S. Water Utilities

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Abstract

Wildfire risk to Western water utility infrastructure is a significant and growing threat, but investment in wildfire risk mitigation programs for water utilities often faces political and public opposition due to associated rate increases. Until now, no centralized database has provided the financial and water quality data needed to quantitatively justify increased investment from water utilities in wildfire risk mitigation programs. To address this gap, we synthesize data from utility financial records, source water area shapefiles, burn severity and wildfire perimeter data, and interviews with water utility decision makers and partners in order to estimate economic impacts on water utilities from individual wildfires, and provide general insights on wildfire impacts for water utilities across the Western United States. These findings are organized into EMBER (Economic and environmental Measurements of Burn Events on water Resources), an interactive database which maps and quantifies the economic and environmental impacts of wildfires on water utilities in the Western United States. Findings displayed in EMBER include the estimate that the Holiday Farm Fire in 2020 had an estimated direct impact of $22.5 million on the Eugene Water & Electric Board, and the visualization of previous findings of an increase in heavy metals up to 16 times the EPA acute criteria in source waters for the Paradise Irrigation District following the Camp Fire in 2018. These case studies and general insights are publicly accessible through the EMBER online tool. EMBER thus serves as a decision support tool for water utility stakeholders and a database for researchers, enabling more informed investment in and management of wildfire mitigation programs for Western water utilities.

1 Introduction

As the Western United States faces hotter summers and reduced snowfall in the winters, wildfires are increasing in frequency and severity (1). Forest management strategies, such as ecological thinning and prescribed burns, can reduce wildfire impacts on forests and the ecosystem services they provide, but financing and implementing these management projects is often limited by political, financial, or regulatory hurdles. Some of the biggest beneficiaries from these projects are water utility companies, for whom wildfires cause expensive degradation to water quality, infrastructure, and reservoir capacity.

It can be more cost-effective, particularly when accounting for co-benefits, for a water utility to invest in fire mitigation practices such as forest thinning or riparian zone restoration than it is for them to pay for post-fire cleanup efforts such as reservoir sedimentation, water quality degradation, and infrastructure damage (2). Yearly costs associated with wildfire impact can reach up to twice a water utility's pre-fire annual operating revenue, as in the case of the Paradise Irrigation District after the 2018 Camp Fire. But despite being some of the most at-risk organizations to wildfire impacts, water utilities face significant barriers to implementing wildfire risk mitigation programs.

To better understand and address the challenges that water utilities face, we conducted interviews with 10 water utility professionals. We found that while the most common challenge was budget restrictions, even water utilities with the necessary budget allocations still faced difficulties implementing wildfire risk mitigation programs because of regulations against raising rates, doubts from board members about return on investment, and understaffing at crucial partner agencies such as the U.S. Forest Service. One interviewee noted that since their agency had not been exposed to wildfire, their board would only be receptive to wildfire mitigation programs with clear cost-benefit analysis, and they didn't have the data to support that analysis. Another interviewee similarly hypothesized that board and ratepayer opposition could be addressed by creating better educational materials about the return on investment of wildfire mitigation programs.

Through these interviews and our own literature review, we identified a crucial gap in available data regarding the economic impact of wildfires on water utilities, and hypothesize that the clear communication of this data would illuminate the benefits of investments in wildfire mitigation programs through data from other utilities who have suffered significant post-fire losses. To test this hypothesis, we created EMBER (Economic and environmental Measurements of Burn Events on water Resources), an interactive database which visualizes the economic impact of wildfires on water utilities across the United States.

EMBER contains geospatial data which presents case study analysis of a number of water utilities in Colorado, California, Washington, and Oregon. It can also be used to query which wildfire burn perimeter overlapped with a given water utility's source or service area or which water utilities overlapped with a given wildfire's burn perimeter, and presents general trends on burned area over time in the entire Western United States, as well as on a state-by-state basis.

2 Methods

2.1 Case-Study Methods

2.1.1 General Methodology

The primary objective of each EMBER case study is to provide a picture of the economic impact of wildfires on a specific water utility. To do so, we compiled a table of costs that were deemed to have been incurred by the water utility as a result of one or multiple past wildfires. Our process for identifying and quantifying such costs will be described in this section. Information sources were prioritized according to the following hierarchy: 1) Official documentation from the water utility itself or associated entities, 2) Grey literature or peer-reviewed publications, 3) News and other third-party articles.

To prevent subjectivity in the identification of wildfire-induced costs, we used a common standard of evidence as a prerequisite for any cost being reported in our database. This standard required that each cost be explicitly attributed to either the wildfire itself or an established effect of the wildfire by source documentation. As a result, instances where the causal link between wildfire and cost was based purely on speculation were not included, and each cost reported in this database is supported by a chain of explicit documentation. This practice defers the burden of determining causation to the authors of the source documents and minimizes additional assumption-making. However, in instances where further assumptions were necessary to make this connection, these were stated clearly in the cost table. No distinction was made between costs that were absolutely necessary for the utility's operation following the fire and costs supporting more optional projects that would have been best practice for the recovery of the utility and its source water. Such a distinction would require a much more in-depth understanding of the utilities' internal situations

during the response to the fire. Finally, preventative efforts to mitigate the impacts of future fires, although potentially motivated by previous fires, were not considered wildfire-induced costs, and were therefore not included in our case-study cost tables.

For the purposes of EMBER, we define a cost as any adverse financial impact borne by the utility and attributable to the wildfire, encompassing both expenditures incurred and revenue foregone. This could include special projects, operating expenses, capital expenditures, or a decline in operating revenue, among other possibilities. Importantly, all forms of costs were subject to the same standard of evidence for inclusion in EMBER. Whenever possible, we reported costs based on their actual amount rather than budgeted plans. If a budgeted cost plan was reported but lacked a later confirmation of the exact expenditure, the assumption was made that the budget reflects the exact expenditure unless there was any evidence that the budgeted plan was not fully executed. This assumption, when made, was reported in the cost table. In the case where a water utility operates as part of a larger entity, such as a municipality or electric utility, we only reported costs that were pertinent to the entity's water utility segment. For example, the Pateros Water System is a part of the broader municipal government that is the City of Pateros. Following the Carlton Complex Fire, the city faced a variety of costs including debris removal and reconstruction projects, but these were not reported here because there was no evidence that these projects were relevant to the operation of the water utility itself.

We reported project-based costs under the fiscal year during which the associated work is believed to have taken place in order to generate a time series of post-fire costs for each case study. Because report documents from different water utilities can operate under different fiscal year timelines, we also indicate this on the EMBER dashboard. Most commonly, utility fiscal years end on the last day of either December or June. For simplicity, we notate fiscal years by the calendar year during which the fiscal year ended (that is, for an expense that took place throughout the FY 2022-2023 ending in June, we report the cost under 2023). Whenever possible, long-term costs were divided up by year. If the cost of a project spanning multiple years could not be precisely divided up, we reported the cost under the range of years within the project's duration, and the total project cost was divided evenly between these years for the time series. However, note that the priority of this study was to quantify the wildfire-induced costs rather than attribute them to each year with certainty. The latter was often made difficult by data availability and varying accounting standards.

All costs in the table are reported in terms of both a raw value and an inflation-adjusted value. All inflation-adjusted values are expressed in terms of January 2026 USD and were computed using the U.S. Bureau of Labor Statistics CPI Inflation Calculator. For simplicity, the raw values of all costs were input into the calculator using the final month of the fiscal year to which we attribute the cost. For long-term costs which could not be divided up by year, the midpoint of the cost's time interval was used as the inflation-adjustment input. For simplicity, all dollar amounts were truncated to the thousands place. All cost figures and totals displayed in EMBER use the inflation-adjusted value.

An estimate for the total economic impact of wildfires on each water utility was calculated by summing the utility's wildfire-induced costs (inflation-adjusted amounts), and the time interval that was studied to reach this estimate is indicated beside it on the EMBER dashboard. The costs included in this total economic impact estimate were only those that seem to be caused entirely by the wildfire. That is, no alternative factors unrelated to the fire were contributing to the cost simultaneously. We defined a parameter for distinguishing between these types of costs, which was termed the "degree of causation" (DoC). It holds a value of 1 for costs that could defensibly be said to be entirely wildfire-induced, and a value of 2 for costs with other unrelated contributing factors, from which the portion of the cost that was caused by wildfires could not be fully disentangled, and without which the cost would not have been the same. For example, Denver Water signed a

contract with the Colorado State Forest Service to implement a vegetation recovery program in the acreage that was burned by the Hayman Fire. However, the contract also implemented work to combat pine beetle issues and establish general forest management practices. The total cost of this contract could not be broken down by project type and was reported with a DoC of 2. One common caveat with this distinction arises when a wildfire leaves the utility's watershed highly susceptible to erosion, which is then carried out by heavy storms that follow. We report any erosion-control costs in this scenario with a DoC of 1, as we consider the burning of the landscape to be the principal cause of the erosion and the heavy storms to be a natural part of the seasonal cycle which would have had only a fraction of the impact without the effects of the wildfire. Costs with a DoC equal to 2 were not included in the economic impact totals, but were still reported in the cost table for completeness.

Water utilities that may have been substantially impacted by wildfires, but for which data tying costs to causation are scarce, may still show null results in our cost quantification efforts. It can be difficult to differentiate between utilities that were not impacted significantly by wildfires and those that simply didn't report the impacts with enough specificity. Confirmation from members of the water utility itself would be required to conclude that a wildfire had little to no financial impact on the company.

Similarly, water utilities for which we successfully conducted case studies may have experienced some further costs that were left unreported or unquantified. Therefore, the list of costs we report here is not intended to be entirely comprehensive, and our estimates of total economic impact can be considered a sort of lower bound. The exclusion of costs with a degree of causation of 2 from total impact estimates also contributes to this notion.

Depending on the available information, we employed a few different methods for quantifying these wildfire-induced costs. To ensure consistency across methods, the same standard of evidence was enforced for each as described above. Each of these methods will be described in the subsections that follow. The table of costs for each case study indicates the method used for the quantification of each cost. In addition, each row of the cost table includes the years in which the costs were incurred, names of the fires that contributed to the cost, links to the relevant source documents, degree of causation, and a description of the cost and all relevant information that was discovered. The description column also reports any unusual assumptions or potential complications that arose during cost quantification.

2.1.2 Direct Quantification Method

When using the direct quantification method, the cost is quantified by the source documents themselves, typically in a narrative format with a concrete dollar amount. This method requires simply recording that dollar amount within the cost table. It involves the least amount of assumption-making, and total wildfire-induced cost estimates that result from this method function as lower bounds with high confidence. As such, the use of this method represents the ideal scenario for cost quantification, although its use is often limited by the specificity of the reporting party. It is most commonly applicable to project-based costs. For example, Denver Water reported that they spent $1.1 million on the construction of a sediment trap along Turkey Creek in 2004. The direct quantification method involves simply pulling that number from the report.

Although the general premise of this method is that minimal calculations are required, at times, simple calculations were conducted to disentangle another explicitly mentioned cost factor or divide the cost between years. When needed, these additional calculations were explained in the notes along with any assumptions made. Tax was included in our costs when it was included by the reporting source, but it was not sought out or calculated without confirmation.

Sometimes, a real cost could not be quantified at all by the available documentation. In this case, all known information was recorded in the description column, while the cost column was populated with "unspecified" or "unquantified."

2.1.3 Baseline Method

Rather than provide specific, itemized costs for wildfire recovery work, water utility financial reports sometimes made more generalized statements that specific categories of costs were elevated due to certain wildfire-induced needs. For example, a company could report that their annual source of supply (SoS) costs were elevated in FY 2021 due to recovery work being done in the burned watershed. In such a situation, we employed the baseline method to calculate the amount by which this category of costs had been elevated due to the wildfire. This involved identifying the pre-fire annual amount spent on that cost category and using that as a baseline for comparison. The first full fiscal year that preceded the onset of the fire was selected as the baseline year unless there was some extraordinary circumstance that prevented it from being considered a good baseline, such as a separate elevated cost in that category during that year. If the year immediately prior was not deemed a proper baseline, then the years before that were considered, and the selection would be indicated in the cost table.

After identifying the baseline year, the amount spent on the cost category of interest during that year was inflation-adjusted to match each of the years in which this cost category was elevated due to the effects of the wildfire. Specifically, the baseline was inflation-adjusted to the final month of the fiscal years of interest. Finally, the inflation-adjusted baseline was subtracted from the elevated amount to yield the amount by which the cost category had risen as a result of the wildfire.

Notably, this relies on the assumption that the only factors contributing to the elevation in the cost category are wildfire-induced. We believe this assumption is reasonable if the financial report explicitly attributes the rise in cost only to the effects of the wildfire. If the report mentioned any alternative cost drivers, the effects of these drivers were either disentangled, or the cost was reported with a degree of causation of 2. Either way, this methodology would be described within the cost table.

Because of this assumption, the baseline method carries a possibility of overestimating. In reality, though, it may actually underestimate the magnitude of many wildfire-induced costs, since the utility may divert some baseline funds from pre-existing projects as part of the wildfire response, meaning that more than just the difference would be put towards wildfire costs. It is also possible that the utility faced certain wildfire-induced costs that do not fall into the cost categories for which the baseline method was conducted. In this case, the direct quantification method could be used in tandem with the baseline method to get a more complete picture. Whenever employing multiple methods within the same case study, care was taken to ensure that costs would not be double counted. If there was ever uncertainty about whether a cost would fall into a particular category, this was flagged in the cost table.

We used the baseline method not only for specific cost categories (such as source of supply costs), but also for broader figures such as annual operating expenses or annual operating revenue. The exact cost category or annual figure that was used for the baseline method in each case study was determined by the wording of the source documentation. For all cases, the same standard of evidence was applied. For example, this means that an operating revenue baseline was only used if the source documents explicitly stated that operating revenue declines were due to the impacts of the wildfire.

When an operating expense baseline was used, we subtracted depreciation from annual operating expenses across the board prior to calculation. This was done for two reasons. First, if a case study

also reported the cost of a wildfire-induced capital expenditure, the depreciation of this capital asset would be double counted by the operating expense baseline in subsequent years. Second, operating expenses net of depreciation provide a better picture of the company's real expenses in each year without the noise from unrelated changes in capital assets.

Dollar amounts used in baseline calculations were drawn from utility financial reports prepared on an accrual basis. We attribute the costs from each year's baseline calculation to the immediate fiscal year, even though the work for some accrued expenses may not take place until the following years. For this reason, there may be a temporal discrepancy between costs quantified using the baseline method and those quantified using the direct quantification method.

2.1.4 Capital Expenditures Method

The capital additions method is aimed at tracking capital expenditures that were demanded in some sense by the effects of a wildfire. Like the direct quantification method, it requires very little calculation, although it requires the additional step of pulling the dollar amounts from a table of capital asset additions.

This method was commonly applicable to situations where the utility experienced significant destruction of its infrastructure due to burning. In such a case, the utility financial reports often provided the value of all destroyed assets, and the costs of replacing them with new capital assets were reported throughout subsequent years. For this method, we only recorded the latter. There is sometimes a large disconnect between the historical value of old, destroyed infrastructure and the cost required to replace it due to changing costs of construction and materials and the changing needs of the utility. Recording the gross costs of new capital expenditures is also consistent with our focus on real post-fire costs.

The capital asset additions that we considered wildfire-induced costs were only those for which there was strong evidence that they were necessitated by the effects of the wildfire. This was informed by a combination of the written narrative and the categories of assets that were destroyed. Any assumptions needed to reach this conclusion were reported in the cost table. These assumptions inherently carry a risk of overestimating wildfire-induced costs if they do not hold. We credited costs derived from this method to the fiscal year during which those capital asset additions were reported by the utility.

2.1.5 Wildfire-Based Aid

All costs and estimates of total economic impact reported in EMBER are gross of any recovery income. Therefore, we separately reported any sources of wildfire-based aid funding that surfaced from our research. These could include federal or state-sponsored grants and reimbursements, insurance proceeds, legal settlements, and more. Notably, we did not include sources of financing such as loans, unless the debt from such loans was forgiven.

To identify and quantify these sources of wildfire-based aid, we followed the same procedure as for costs. Thus, only sources of aid that were explicitly said to be in response to the wildfire or effects of the wildfire were included. We also only reported aid that originated from sources external to the utility. This means that internally generated revenue in response to the wildfire, such as that from increasing rates or service fees, would not be included in aid tables and totals. Sources of aid were also subject to a degree of causation of 2 if only a portion of the funding went towards wildfire-induced costs.

Sources of aid were summed to yield the total amount of wildfire-based aid that the utility received during the time interval studied. In theory, all costs covered by wildfire-based aid would also

be reported separately in the cost table. However, the exact projects toward which grant funding was allocated were scarcely reported, so this can be difficult to verify. Additionally, sources of wildfire-based aid are typically reported with more specificity and comprehensiveness than wildfire-induced costs. As a result, total estimates of wildfire-based aid may be less likely to underestimate the actual total than our total estimates of wildfire-induced costs. In some cases, the total wildfire-based aid even exceeded our total cost estimate.

Individual sources of wildfire-based aid were also organized by the year in which they were awarded, and a time series plot was generated to parallel the plot for costs. However, because there can be a temporal discrepancy between the year aid was awarded and the year the project took place, costs and aid amounts within a given year may not be directly comparable, and net cost figures should be computed over the full study period rather than year by year.

2.1.6 Pre-Fire Annual Operating Revenue

The sum of costs that a water utility faced following a wildfire could carry a dramatically different financial weight depending on the utility's size. In an attempt to provide a rough benchmark for the utility's operating size and ability to absorb sudden costs, we reported the utility's pre-fire annual revenue using two different metrics: annual water revenue and annual revenue of the entire entity. The pre-fire annual water revenue is the company's revenue derived exclusively from water service business, such as service fees and quantity charges. This excludes any investment earnings, taxes, grants, or other revenue from any non-water business segments such as sewer, electricity, or recreation. On the other hand, the pre-fire annual total revenue for the entire entity is the company's entire revenue inclusive of everything except for debt proceeds, bond premiums, interfund transfers, beginning balances, and asset sale proceeds.

These numbers were typically pulled from the first full fiscal year that preceded the onset of the most impactful wildfire that the utility faced and were inflation-adjusted to be expressed in January 2026 USD. The exact fiscal year that was used is indicated on the EMBER dashboard. These two metrics are intended to function as rough proxies for the operating size of the utility's water segment and company as a whole, respectively, at the time the fire occurred. They are not intended to be precise measurements of the company's vulnerability to costs, as there are many additional factors that must be considered to do so. For companies that specialize purely in water sales and services, these two numbers are often very similar in magnitude, while they can be dramatically different for utilities that operate as part of a larger government or corporation.

2.2 Geospatial Data Methods

We preprocessed all data used in EMBER into Parquet tables, allowing for more efficiency querying during live use of the dashboard. A data ingestion pipeline uses Python and SQL to process geospatial water area source and service area boundaries, wildfire perimeters, and burn severity rasters into organized tables stored in Google Cloud Storage, which EMBER accesses through a read-only service account.

2.2.1 Water Utility Source Areas

Of the four states analyzed in EMBER, only Oregon and Washington had publicly available datasets for their water utility's source water areas, defined as the watershed providing water to the utility which then gets served to customers. For Oregon, we used the State of Oregon Surface Water Drinking Water Source Areas dataset (3). The features are public water system source-area polygons,

which we combined into one polygon per utility so that a utility with multiple intakes appears once in EMBER as a unified entry.

For Washington, we used the Washington Department of Health source-water protection areas (4). EMBER uses the full watershed file provided by the Department of Health, removing inactive systems and the utilities not listed as having their "full" watershed provided. Similarly to Oregon, we combine into one geometry per water system.

For the Denver Water case study, we manually constructed the source watershed using publicly available source water area information from Denver Water and HUC-10 level watershed shapefiles from the state of Colorado (5)(6).

2.2.2 Water Utility Service Areas

After reaching out to professionals at the California Department of Water Resources, we concluded that no similar source water shapefiles exist for water utility source areas in California. Instead, we used the California Water District Boundaries dataset (7), which gives the boundaries for all public water agencies in California including public water systems, agricultural water districts, urban water districts, Federal and State water contractors, wholesalers, retailers, and other public or private utilities that deliver water to the end user.

3 Results and Discussion

3.1 Case Study Results

We have conducted eight case studies to date, following the methods outlined above. These included seven water utilities and one electric utility. Collectively these eight case studies quantified a total of 255 million dollars of wildfire-induced costs. Table 1 shows key economic metrics for each of these case studies and the key cost contributors. Itemized cost breakdowns, detailed descriptions, and sources can be found in the case study view of the EMBER dashboard.

For water utilities, the greatest costs associated with each wildfire commonly did not occur immediately after the fire, but a few years later. We found that the year with the greatest total wildfire-induced costs occurred an average of X years after the year of the fire. This seems to be because, during the years that immediately follow the fire, the utility's main priority is resuming operations to the best of their ability and providing workable fixes to the most pressing issues. The most costly projects are often those that have the company's long-term interest in mind, such as sediment dredging or process adaptations. Costly projects also may require a significant amount of time for internal approval and the approval of external funding such as federal grants, further contributing to this observed delay.

Throughout the process of the case-study research, we found that water utilities that draw from reservoirs or lakes tend to be most vulnerable to water resource-related costs, while those that draw from rivers are the second most vulnerable, and those that draw from groundwater are the least. This seems to be because reservoirs and lakes have the highest potential to be permanently altered by the effects of a wildfire. Rivers, on the other hand, may suffer significant losses in water quality, but these impacts are much less likely to persist for decades. Once sediment is washed into a reservoir, it will remain there permanently until removed. Furthermore, the entity that actually owns the reservoir is typically the entity that incurs the majority of the related costs. Utilities in all three of these categories are susceptible to costs related to physical infrastructure damage from wildfires.

Utility Name	Total Wildfire-Induced Cost	Pre-Fire Annual Water Revenue	Pre-Fire Annual Total Revenue for Entity	Largest Cost Driver
Denver Water	$54,774,000	$267,941,000	$303,892,000	Sediment dredging
Eugene Water & Electric Board (Water)	$22,144,000	$48,213,000	$396,178,000	Erosion control and watershed restoration
Eugene Water & Electric Board (Electric)	$6,243,000	$333,308,000	$396,178,000	Infrastructure damage
Paradise Irrigation District	$94,596,000	$10,593,000	$11,216,000	Infrastructure damage
Casitas Municipal Water District	$2,806,000	$11,591,000	$20,603,000	Sediment removal and infrastructure damage
Pateros Water System	$2,651,000	$302,000	$2,018,000	Infrastructure damage
Central Utah Water Conservancy District	$39,991,000	$33,681,000	$189,987,000	Water quality and turbidity treatment costs
City of Alamogordo Water and Sewer	$31,958,000	$8,077,000	$67,019,000	Sediment dredging

Table 1: Metrics for total wildfire-driven economic cost to utilities. All dollar amounts are reported in January 2026 USD.

3.2 General Quantitative Insights

We present some preliminary general insights from the geospatial data gathered in EMBER in Figure 1, which shows the overlap acreage between wildfires and water utility source areas in Oregon and Washington over the past 45 years. We show a general positive increase in the 5-year rolling average over time until a peak in 2020, after which the average decreases due to the less intense wildfire seasons of 2022 - 2024 in the Pacific Northwest. This intersection is not calculated for California or Colorado, since it is not meaningful without access to the source water area shapefiles for those states.

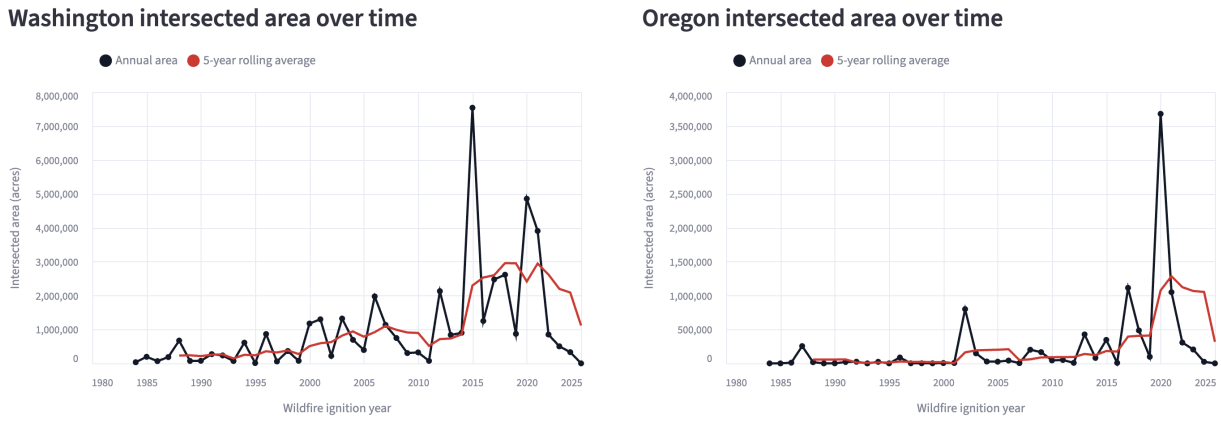


Figure 1: Total overlapped area between utility source water area boundaries and wildfire burn perimeters since 1980.

3.3 Interview Feedback

We provided a demonstration of EMBER's capabilities to our interviewees, receiving positive feedback and constructive criticism. A common trend among interviewees was to remark that the case studies were the most valuable asset in EMBER. One water resources engineer, after being presented with our estimate for economic wildfire-induced damages at the water utility at which they work, noted that different employees at the water utility might have different ideas about which costs should be attributed to wildfire impacts. For example, they pointed to post-fire riparian restoration efforts which served to reduce damage from the wildfire itself, but also act as a preventative measure for further damage. They noted that the preventative portion of the project might not be included in wildfire impact estimates by all parties, though they did acknowledge that it was implemented due to the recognition that it would decrease future wildfire economic damages. They also validated the number we presented, affirming that at least in this case, our assumptions followed industry standards and best practices for tabulating the economic damages that a utility faced due to a given wildfire. In general, interviewees reacted postively to EMBER, and more than one asked if they would be able to access it for use in their own work.

4 Conclusions

EMBER has the potential to address some of the largest barriers that water utilities face when financing and implementing wildfire risk mitigation programs. The case studies provide a reference for the economic impacts that wildfires have had on water utilities, which opens the possibility for future work to study how wildfire risk mitigation efforts could have lessened these costs and to ultimately quantify the return on investment for forest management practices. EMBER increases public awareness of wildfire risk in areas which have not yet been impacted by wildfires by providing visualizations of historic wildfire burn perimeter intersections with nearby utilities of interest, serving as a cautionary tale for those in similar, though yet unaffected, geographic regions.

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